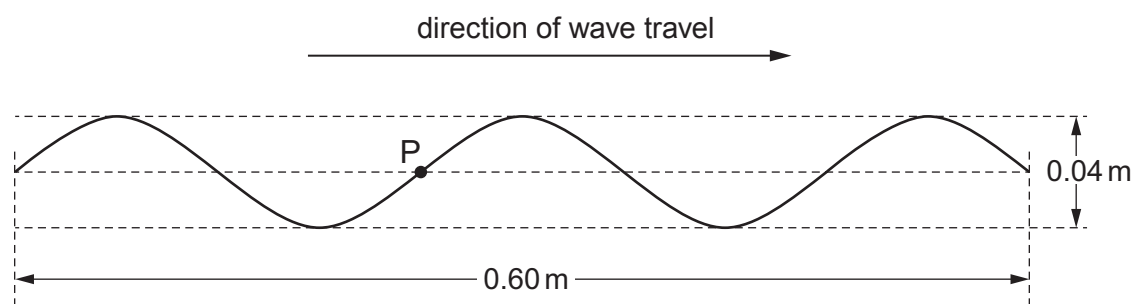
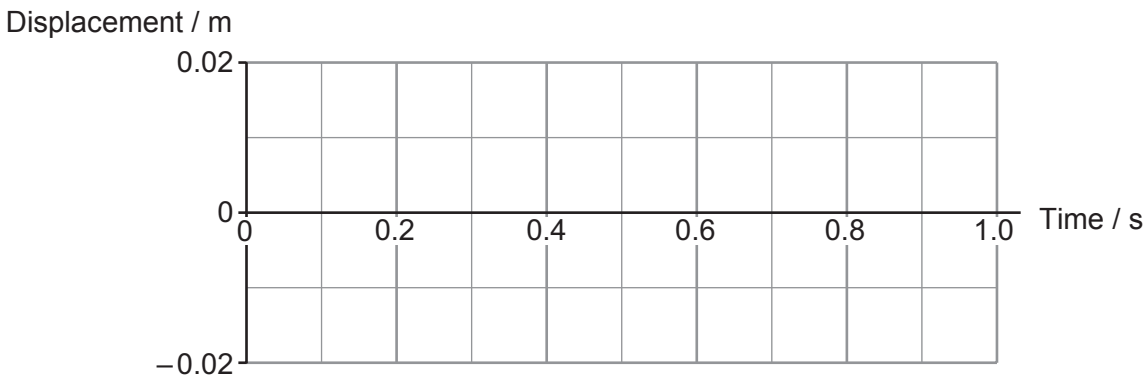


4. (a) A progressive wave is travelling from left to right along a stretched string at a speed of 0.40 m s^{-1} . The diagram shows the string at time $t = 0$.



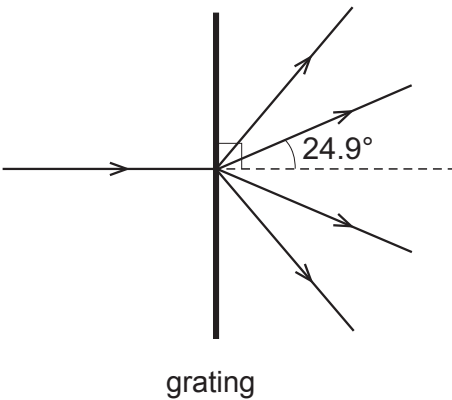
Carefully sketch, **on the grid below**, a displacement–time graph for point **P** on the string between $t = 0$ and $t = 1 \text{ s}$. The space below the grid is for your working. [3]



Space for working:



- (b) The distance between the centres of the slits in a diffraction grating is 1500 nm. Monochromatic light is shone normally on to the grating.



- (i) First order beams emerge at angles of 24.9° to the normal (see diagram). Calculate the wavelength of the light. [2]

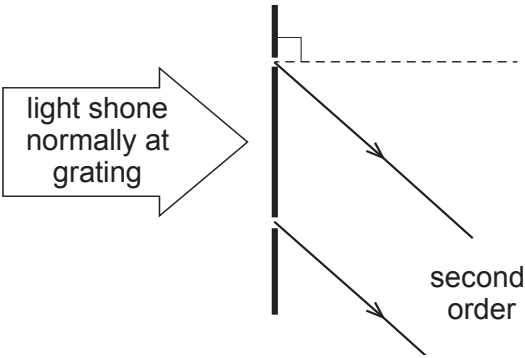
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- (ii) Explain **in terms of path difference** why the **second** order beams emerge from the diffraction grating at 57.4° to the normal. You will need to add to the diagram (which shows two adjacent slits in the grating). [3]



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